

Yuchen Wang

Agentic post-training | ML systems

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Education

University of Illinois Urbana-Champaign	2025 – 2027 (expected)
Master of Computer Science	Champaign, IL
Peking University	2021 – 2025
B.S. in Intelligent Science and Technology (AI); Zhi Class (Elite Program), Top 10%	Beijing, China

Internships

Alibaba U.S. – Accio Team	May 2026 – Dec 2026
Research Intern, LLM Training Algorithms & Infrastructure	Sunnyvale, CA
- Co-developed core RL post-training algorithms and engineered execution infrastructure for Occamy-1.0, a 35B-A3B agent: captured exact model-boundary token IDs and verifier outcomes across OpenClaw, Hermes, and Accio Work. - Recovered long-horizon SFT/RL supervision through workspace replay and context–action–reward realignment after tool failures and compaction, keeping trajectories usable for post-training. - Shipped Occamy with Accio; versus its Qwen3.6 base, the released model raised AutomationBench strict pass 7.5% → 27.6% and Claw-Eval average 69.5 → 82.2, while reducing Claw-Eval T/C full-trial time 36.6%.	
Freedo Technology	May 2025 – Aug 2025
Research Intern, 3D Vision and Generative AI	Beijing, China
Raised point-cloud geometric fidelity from 32.7% to 85.3% by conditioning ControlNet on depth and surface normals while preserving edges and planar surfaces.	

Open Source

- Eliminated a 120B MoE loading OOM path in **vLLM #54699**, halving TP2 BF16 conversion peak **7.88 → 3.94 GiB**; restored calibrated FP8 KV prefill correctness in **FlashInfer #4984**.
- Implemented a Ray-driver bypass in **NeMo RL #3943**, accelerating controlled top-*k* teacher transfers **4.44–5.31×** on 8×B200.
- Implemented **GDN kernel-fused Q/K normalization** and **selective Mamba recompute** for Megatron-LM; the latter ran 24K BF16 after baseline OOM.
- Fixed silent small-gradient failure in **Muon #230**; co-authored **NeMo Gym #2726** to preserve HTTP errors; addressed **tool-schema coercion** and **SSH reconnect** in Hermes.

Research

RareDx: Graph-Grounded RL for Rare-Disease Diagnosis [arXiv] (Under review)	Jun 2026 – Sep 2026
Advisor: Prof. Tianyu Liu, Tsinghua University	
Co-designed RL rewards for partial diagnostic matches using disease-ontology distance, biomedical similarity, and phenotype–disease links. With SFT and retrieval, 27B diagnosis Hit@10 increased from 28.53% to 40.76%.	
CineFlow: Dependency-Driven Parallel Video Generation [Paper] (Under review)	Oct 2025 – May 2026
Advisor: Prof. Fan Lai, University of Illinois Urbana-Champaign	
- Developed a training-free, critical-path scheduler for video diffusion using semantic DAGs, trajectory fusion, and online edge pruning; analyzed 1K+ MovieGenBench prompts. - Delivered 1.7–5.5× end-to-end speedup, 1.30–2.02× lower P90 latency, and 5.4–17.3% higher VBench overall across Wan2.2-5B, CogVideoX-5B, and HunyuanVideo on 8×H100.	
Dynamic Prefill Optimization via Adaptive Online Packing [Report]	Aug 2025 – Jan 2026
University of Illinois Urbana-Champaign	
Built an AIMD batch-size controller driven by live p95 TTFT and packed mixed-length prompts with greedy/DP scheduling; cut TTFT by up to 20% on production-style DynamoLLM traces.	
FlashAttention-style CUDA Kernels for GPT-2 [Report]	Aug 2025 – Dec 2025
University of Illinois Urbana-Champaign	
Accelerated GPT-2 through tiled attention, online softmax, and kernel fusion, raising arithmetic intensity from 15 to 157 FLOP/byte, cutting HBM traffic roughly 10×, and improving end-to-end speed up to 9%.	
Reinforcement Learning for Legal Reasoning [Thesis]	Jan 2025 – Mar 2025
Advisor: Prof. Yansong Feng, Peking University	
Post-trained a legal model with Zero-RL, chain-of-thought-distilled SFT, and GRPO; reached 57.6% accuracy, beating SFT-only and RL-only baselines.	

Awards & Technical Skills

Awards: Merit Student; Zhi Class Scholarship; CMO (Provincial First Prize), 2020.

Languages: Python, C++, CUDA. **Systems:** PyTorch, Ray, FSDP, Megatron-LM, NeMo RL, vLLM, SGLang, Nsight.

Methods: RL, distributed training, diffusion transformers, ControlNet, 3D vision and point clouds.